Active Directory Home Lab – Build Log

Sep 28, 2026 · @zac

Overview

I built a small company IT environment from scratch in VirtualBox: an Active Directory domain, `lab.local`, a domain-joined Windows 11 workstation, and an Ubuntu Linux server running the osTicket help desk. On top of it I worked ten realistic help desk tickets end to end, from password resets and onboarding to Group Policy, DNS and PowerShell automation.

The goal is hands-on practice with the work an IT support or junior sysadmin role handles daily, documented with the reasoning behind each decision and the problems I hit along the way. The roadmap at the end shows what comes next.

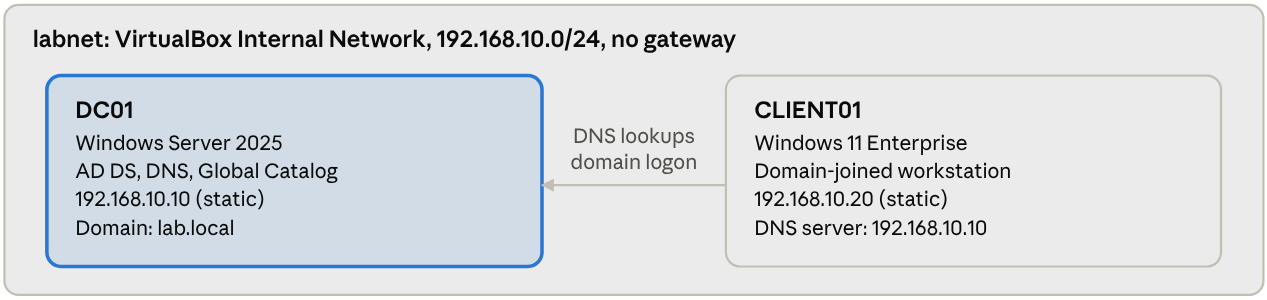
Skills demonstrated

- Windows Server 2025: AD DS, DNS (forward and reverse zones), Group Policy
- User lifecycle: onboarding, offboarding, password resets, account lockouts
- Access control: OUs, security groups, AGDLP, NTFS and share permissions
- PowerShell: bulk user creation from CSV, auditing and fixing account data
- Security practice: least privilege, named admin accounts, UAC, lockout policy
- Linux: Ubuntu Server administration over SSH, packages, services, permissions
- Web stack: Apache, MariaDB and PHP (LAMP) hosting the osTicket help desk
- Troubleshooting: documented problems, causes and fixes throughout

Lab environment

Three VMs share an isolated VirtualBox internal network, `labnet`, with no router. DC01 provides AD and DNS; CLIENT01 and HELPDESK01 use it as their DNS server. HELPDESK01 also has an outbound-only NAT adapter for software downloads.

CLIENT01 finds the domain through DC01 on an isolated network



lab network · 2 VMs on labnet

Every DNS lookup and login from CLIENT01 goes to DC01, which is why the client can't join or log in while DC01 is off.

Item	Value
Host	Laptop, 16 GB RAM
Hypervisor	Oracle VirtualBox
Network	Internal Network `labnet`, 192.168.10.0/24, no gateway
Domain	`lab.local` (NetBIOS name `LAB`), single forest, single domain

VM	OS	Role	vCPU	RAM	Disk	IP address	DNS server
DC01	Windows Server 2025 Standard Evaluation (Desktop Experience)	Domain controller, DNS, Global Catalog, file server	2	4 GB	60 GB	192.168.10.10 (static)	127.0.0.1
CLIENT01	Windows 11 Enterprise Evaluation	Domain-joined workstation	2	4 GB	64 GB	192.168.10.20 (static)	192.168.10.10
HELPDESK01	Ubuntu Server 26.04 LTS	osTicket help desk (Apache, MariaDB, PHP 8.4)	2	2 GB	25 GB	192.168.10.30 (static)	192.168.10.10

Directory structure

OU path	Contents
Company\Users\IT	Alex Lee (`alee`)
Company\Users\HR	Jane Smith (`jsmith`)
Company\Users\Sales	John Brown (`jbrown`)
Company\Groups	`IT-Staff` , `HR-Staff` , `Sales-Staff` (Global, Security)
Company\Computers	CLIENT01

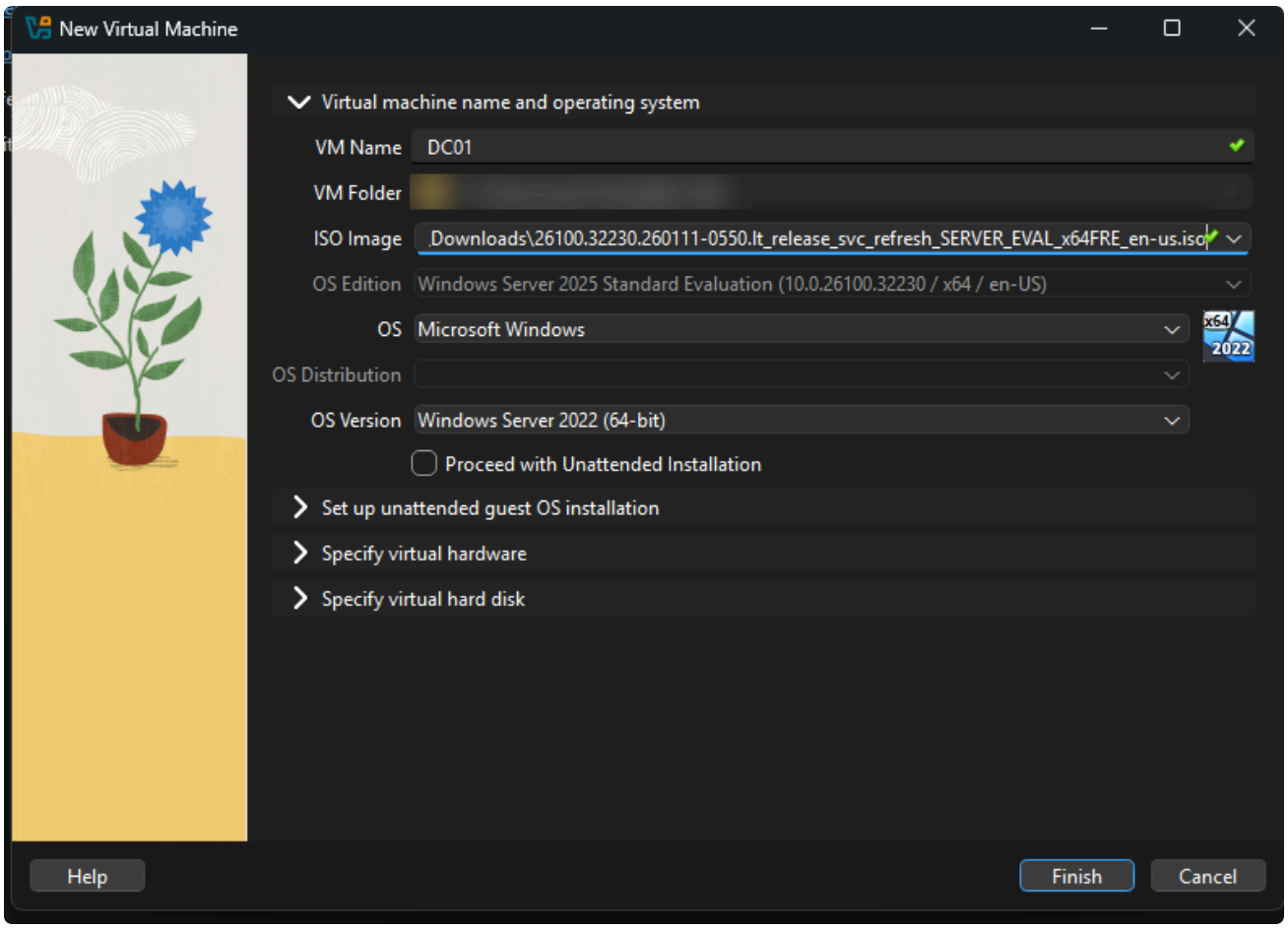
Users and computers sit in separate OUs so Group Policy can target each independently. Each user is a member of their department's group.

Build steps

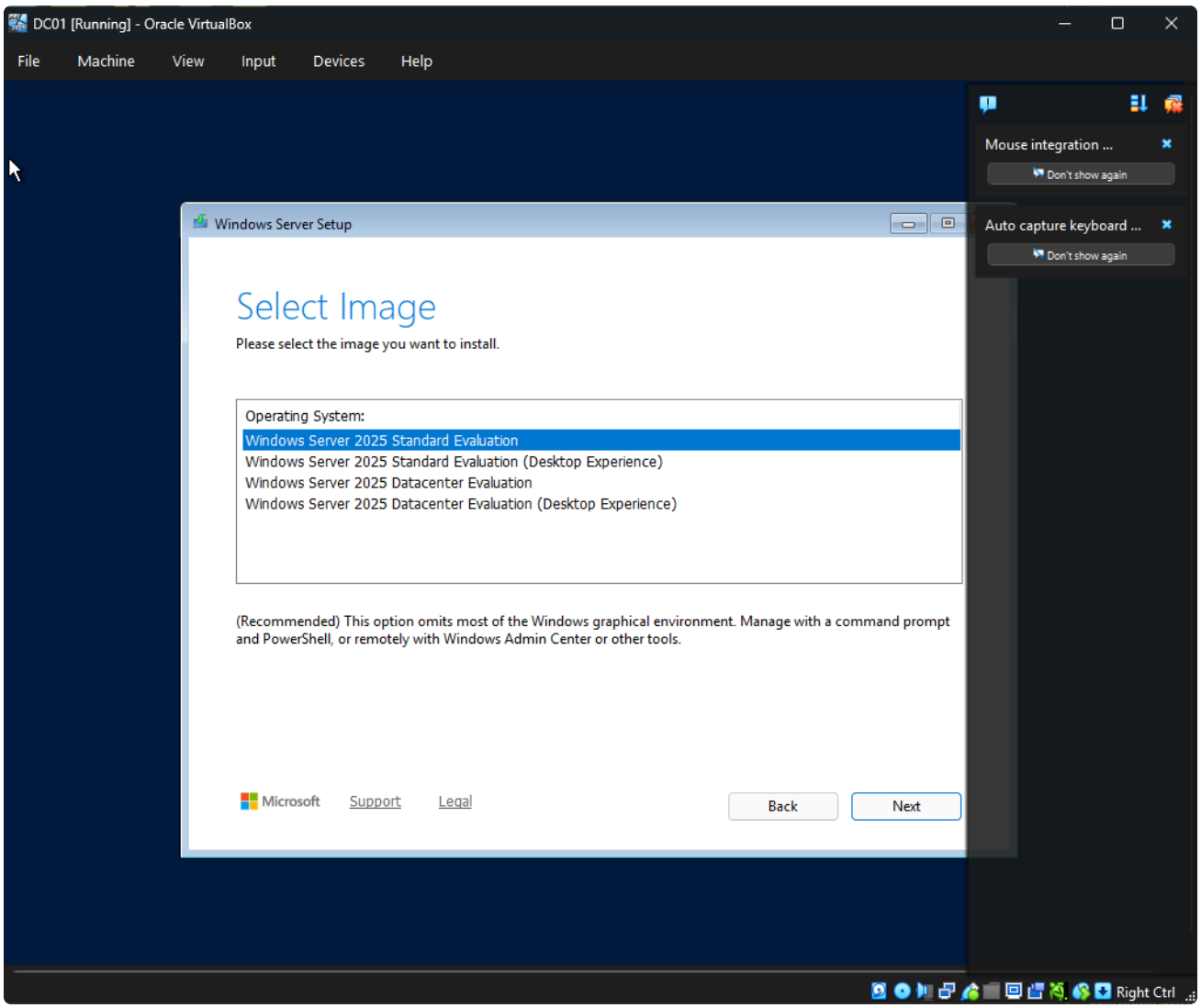
The build took 12 steps, from an empty VirtualBox install to a domain user logged in on CLIENT01. Usernames in file paths are blurred.

Domain controller (DC01)

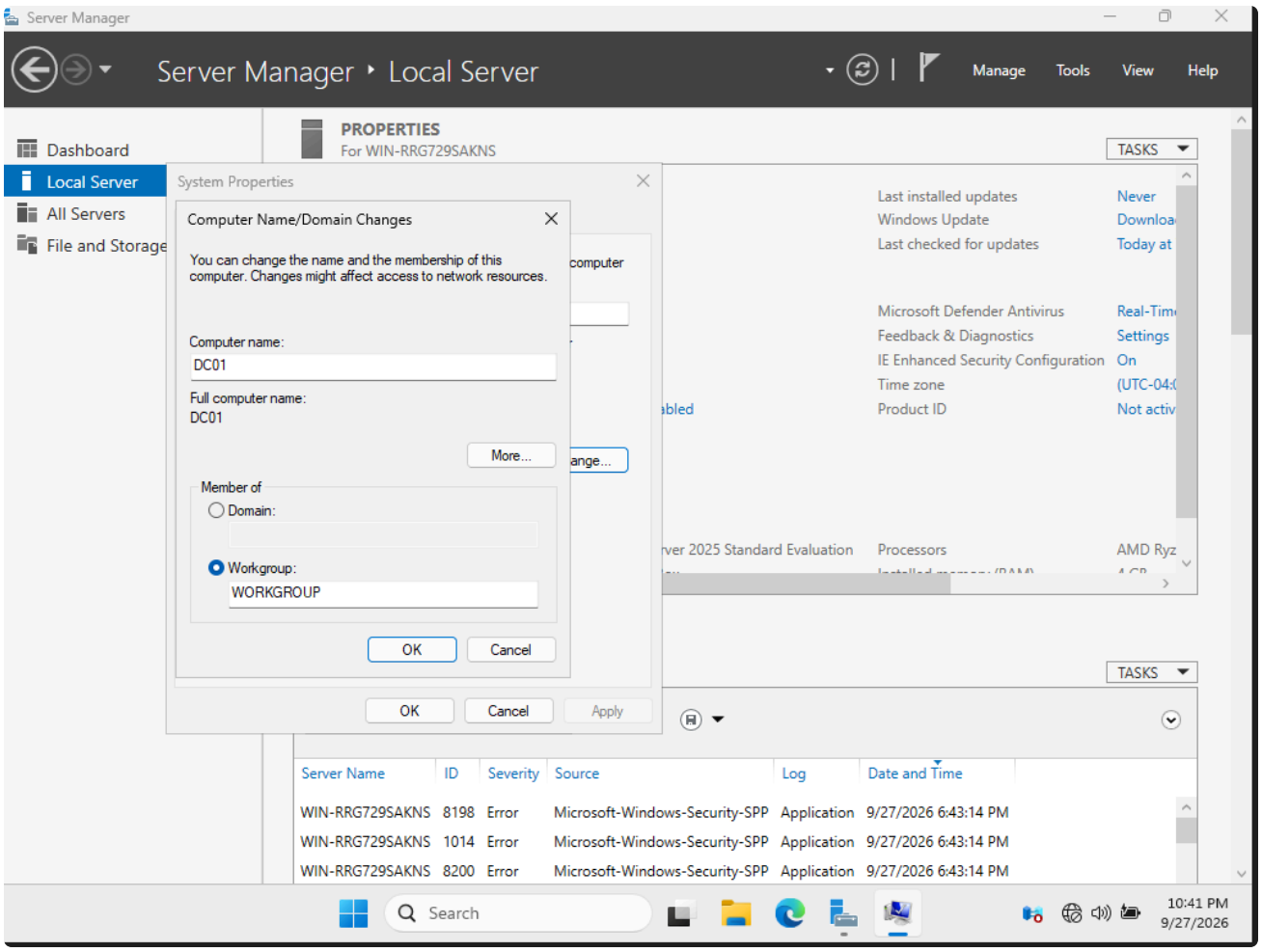
1. **Prepared the host.** Confirmed virtualization was enabled, installed VirtualBox, and downloaded the Windows Server 2025 and Windows 11 Enterprise evaluation ISOs from the Microsoft Evaluation Center.
2. **Created the DC01 VM.** 2 vCPU, 4 GB RAM, 60 GB disk, unattended installation skipped. Attached Adapter 1 to Internal Network `labnet` .

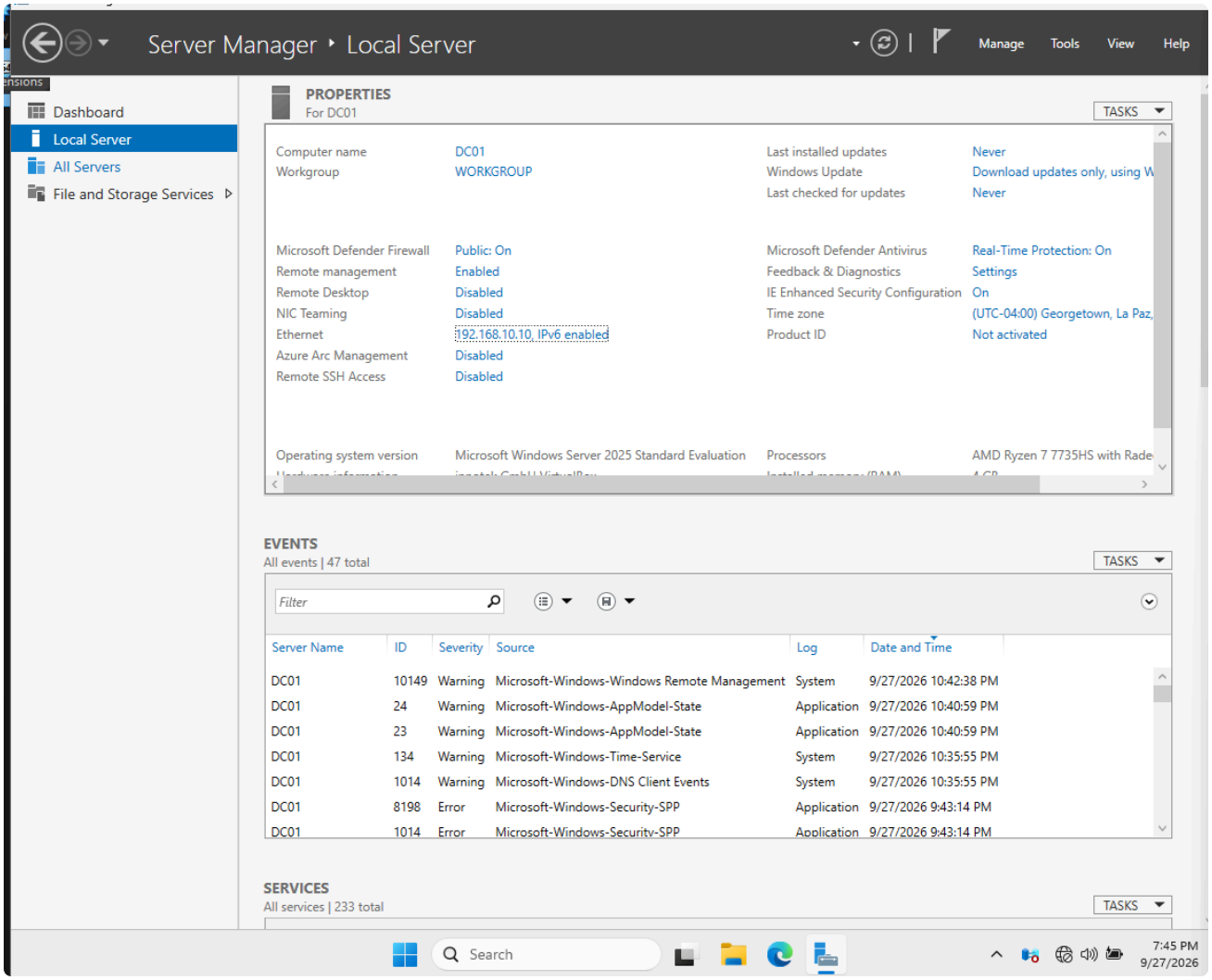


3. **Installed Windows Server 2025.** Chose *Standard Evaluation (Desktop Experience)* for the full GUI, and set the Administrator password.

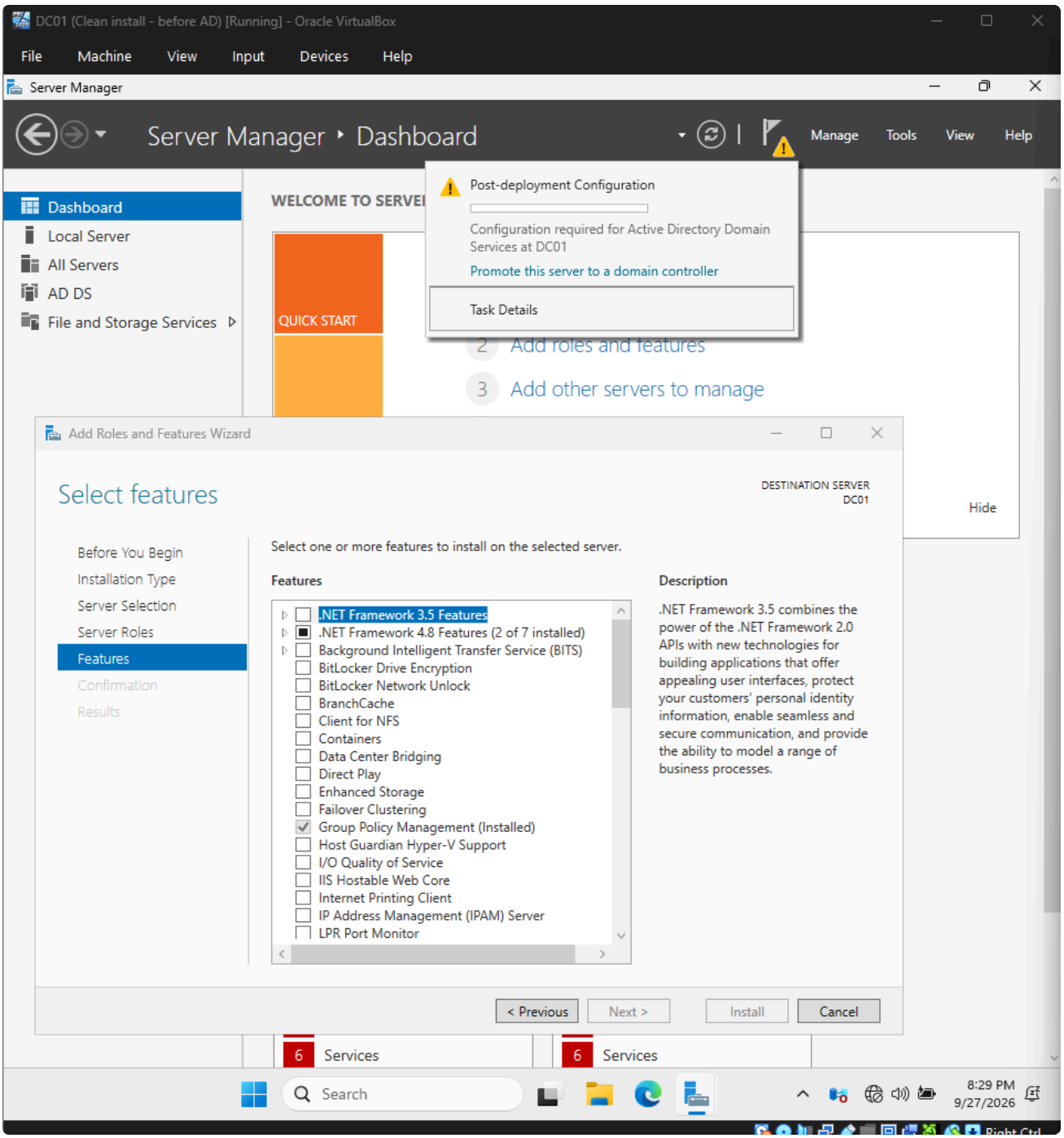


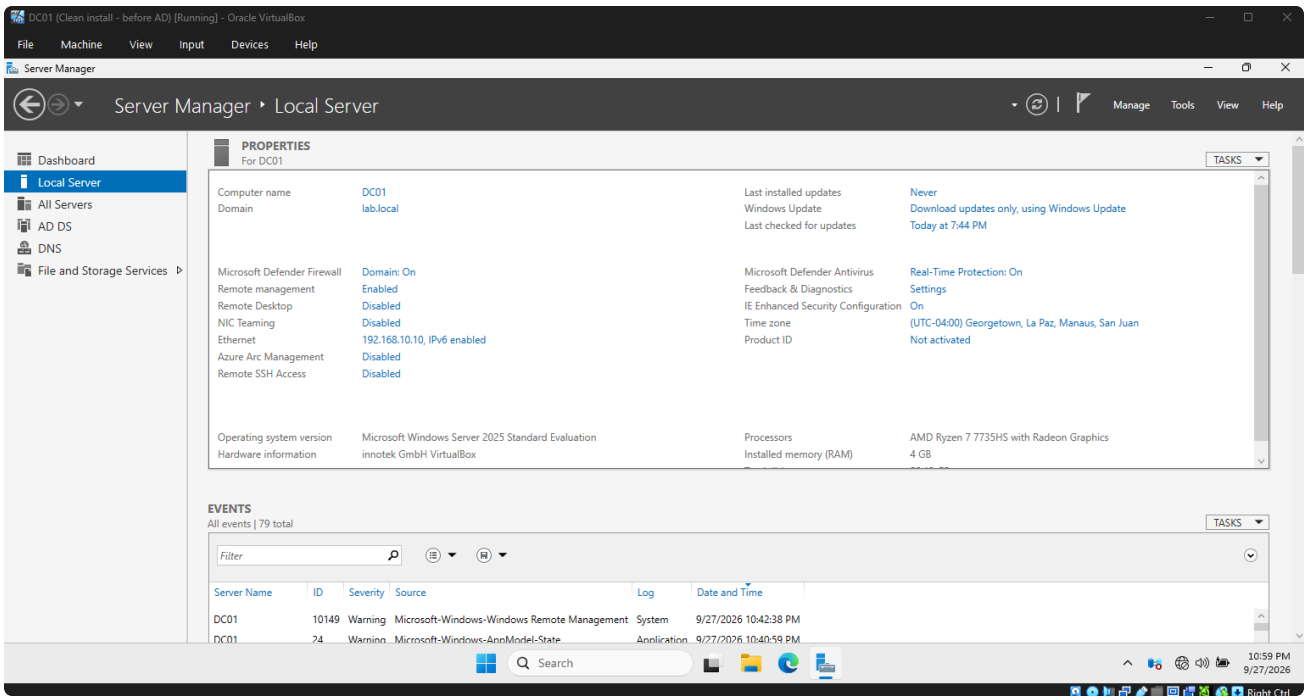
4. **Configured the base server.** Installed VirtualBox Guest Additions, set a static IP (192.168.10.10/24, DNS 127.0.0.1), set the time zone to UTC-04:00, and renamed the server to DC01 before promotion.


Server Manager
Server Manager • Local Server
Manage
Tools
View
Help
Dashboard
Local Server
All Servers
File and Storage
PROPERTIES
For WIN-RRG729SAKNS
TASKS
System Properties
Computer Name/Domain Changes
You can change the name and the membership of this computer. Changes might affect access to network resources.
Computer name:
DC01
Full computer name:
DC01
More...
Member of
Domain:
Workgroup:
WORKGROUP
OK
Cancel
Apply
Last installed updates
Never
Windows Update
Last checked for updates
Today at
Microsoft Defender Antivirus
Feedback & Diagnostics
Settings
IE Enhanced Security Configuration
On
Time zone
Product ID
Processors
AMD Ryz
TASKS
Server Name
ID
Severity
Source
Log
Date and Time
WIN-RRG729SAKNS 8198 Error Microsoft-Windows-Security-SPP Application 9/27/2026 6:43:14 PM
WIN-RRG729SAKNS 1014 Error Microsoft-Windows-Security-SPP Application 9/27/2026 6:43:14 PM
WIN-RRG729SAKNS 8200 Error Microsoft-Windows-Security-SPP Application 9/27/2026 6:43:14 PM
Search
10:41 PM
9/27/2026

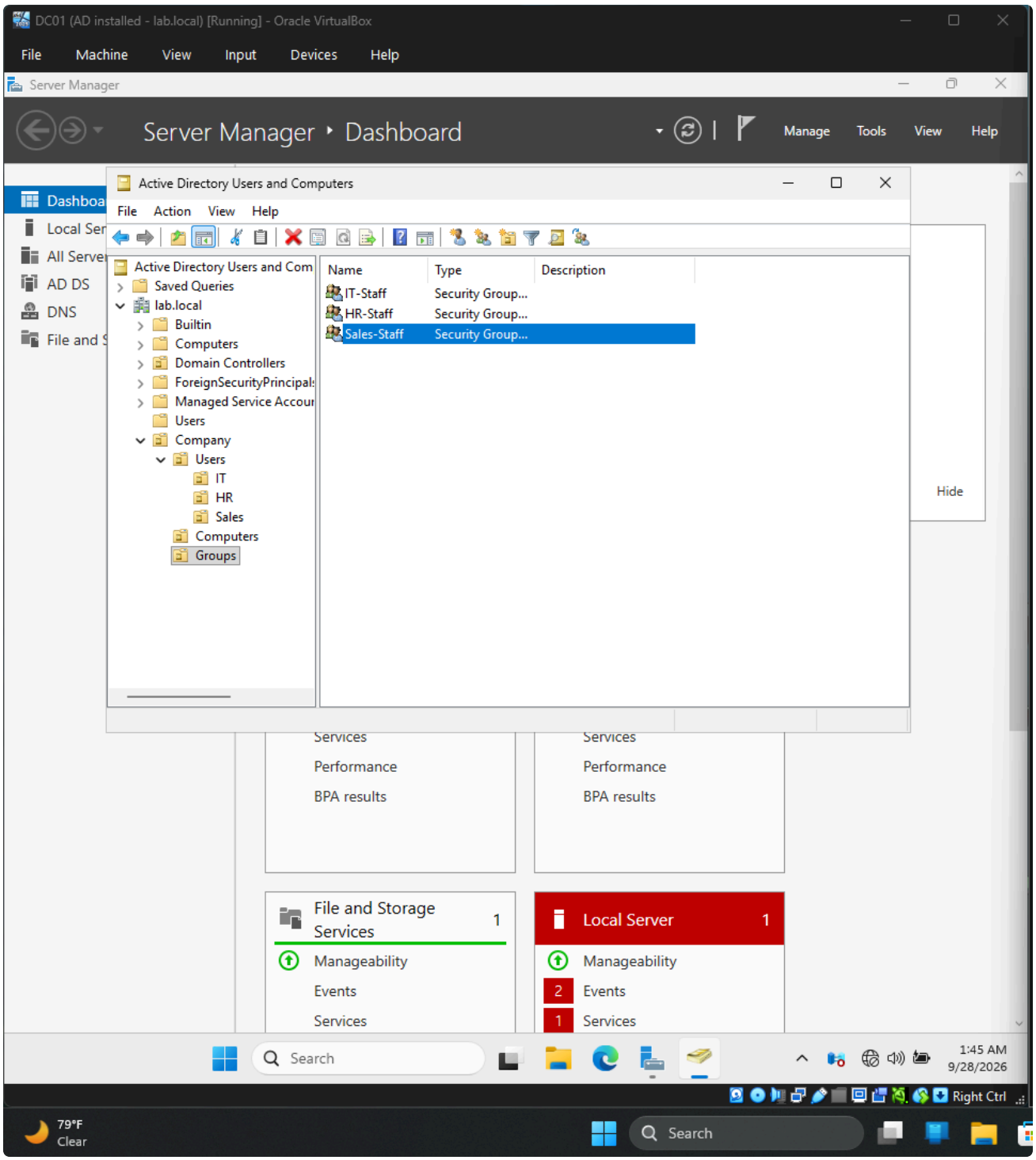


5. **Snapshot:** `Clean install - before AD`.
6. **Installed AD DS and promoted DC01.** Added the Active Directory Domain Services role, then promoted DC01 to a domain controller in a new forest, `lab.local`, with DNS and Global Catalog. NetBIOS name `LAB`. The server rebooted automatically.


DC01 (Clean install - before AD) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Server Manager
Server Manager ▸ Dashboard
Manage Tools View Help
Dashboard
Local Server
All Servers
AD DS
File and Storage Services
WELCOME TO SERVE
QUICK START
Post-deployment Configuration
Configuration required for Active Directory Domain Services at DC01
Promote this server to a domain controller
Task Details
2 Add roles and features
3 Add other servers to manage
Hide
Add Roles and Features Wizard
Select features
DESTINATION SERVER
DC01
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results
Select one or more features to install on the selected server.
Features
.NET Framework 3.5 Features
.NET Framework 4.8 Features (2 of 7 installed)
Background Intelligent Transfer Service (BITS)
BitLocker Drive Encryption
BitLocker Network Unlock
BranchCache
Client for NFS
Containers
Data Center Bridging
Direct Play
Enhanced Storage
Failover Clustering
Group Policy Management (Installed)
Host Guardian Hyper-V Support
I/O Quality of Service
IIS Hostable Web Core
Internet Printing Client
IP Address Management (IPAM) Server
LPR Port Monitor
Description
.NET Framework 3.5 combines the power of the .NET Framework 2.0 APIs with new technologies for building applications that offer appealing user interfaces, protect your customers' personal identity information, enable seamless and secure communication, and provide the ability to model a range of business processes.
< Previous
Next >
Install
Cancel
6 Services
6 Services
Search
8:29 PM
9/27/2026

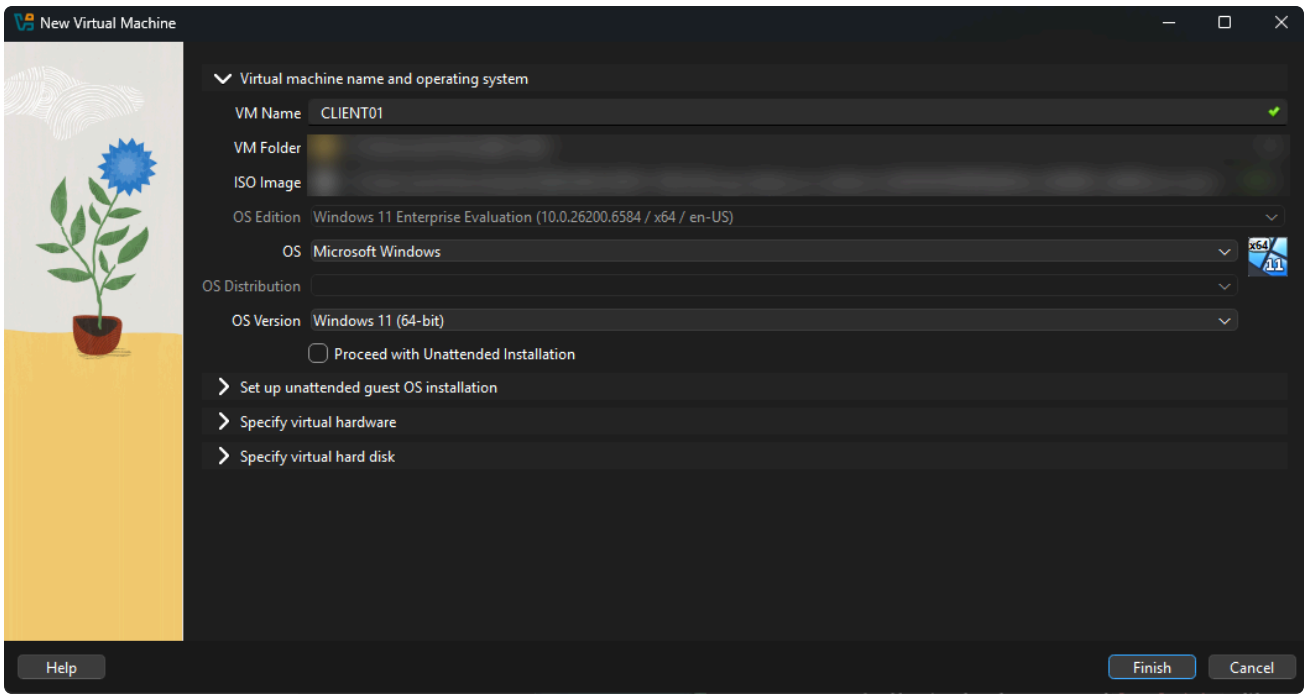


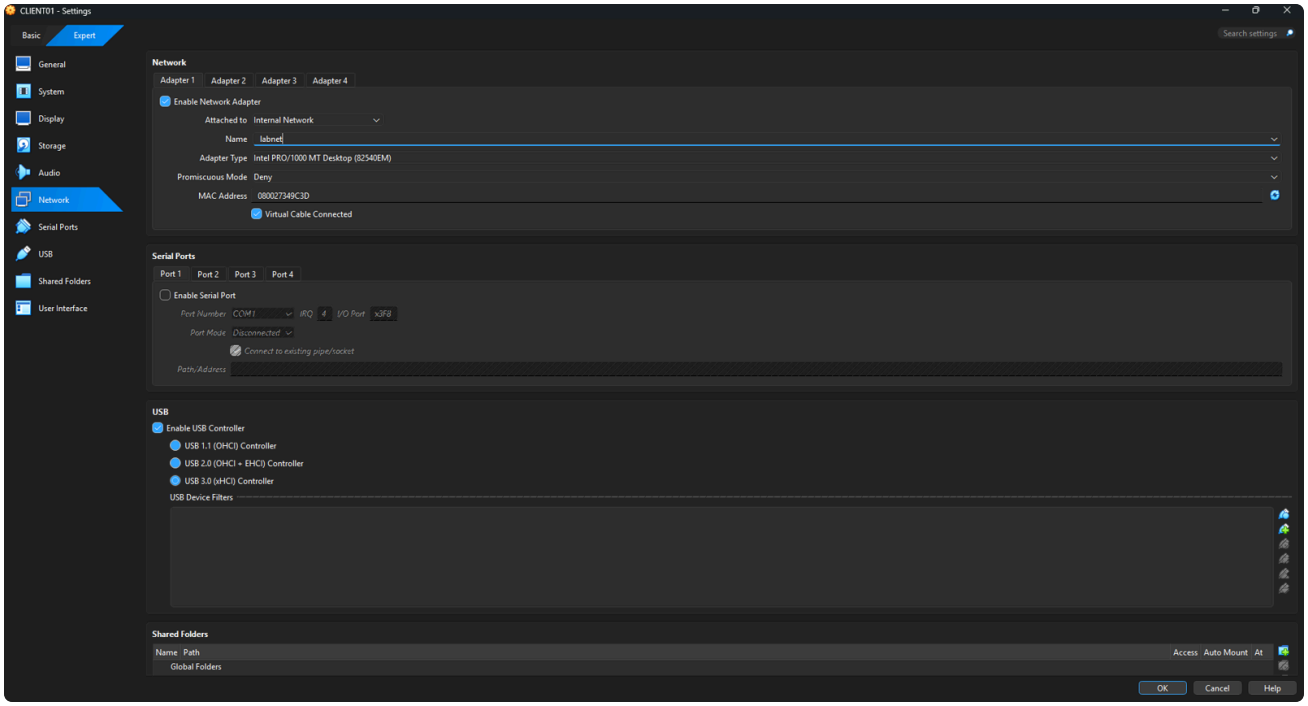
7. **Snapshot:** `AD installed - lab.local`.
8. **Built the OU structure, users and groups** in Active Directory Users and Computers, as listed under Lab environment. Added each user to their department group.



Client (CLIENT01)

9. **Created the CLIENT01 VM.** 2 vCPU, 4 GB RAM, 64 GB disk, OS version Windows 11 (64-bit), which enables the virtual TPM 2.0, EFI and Secure Boot. Attached to `labnet`.

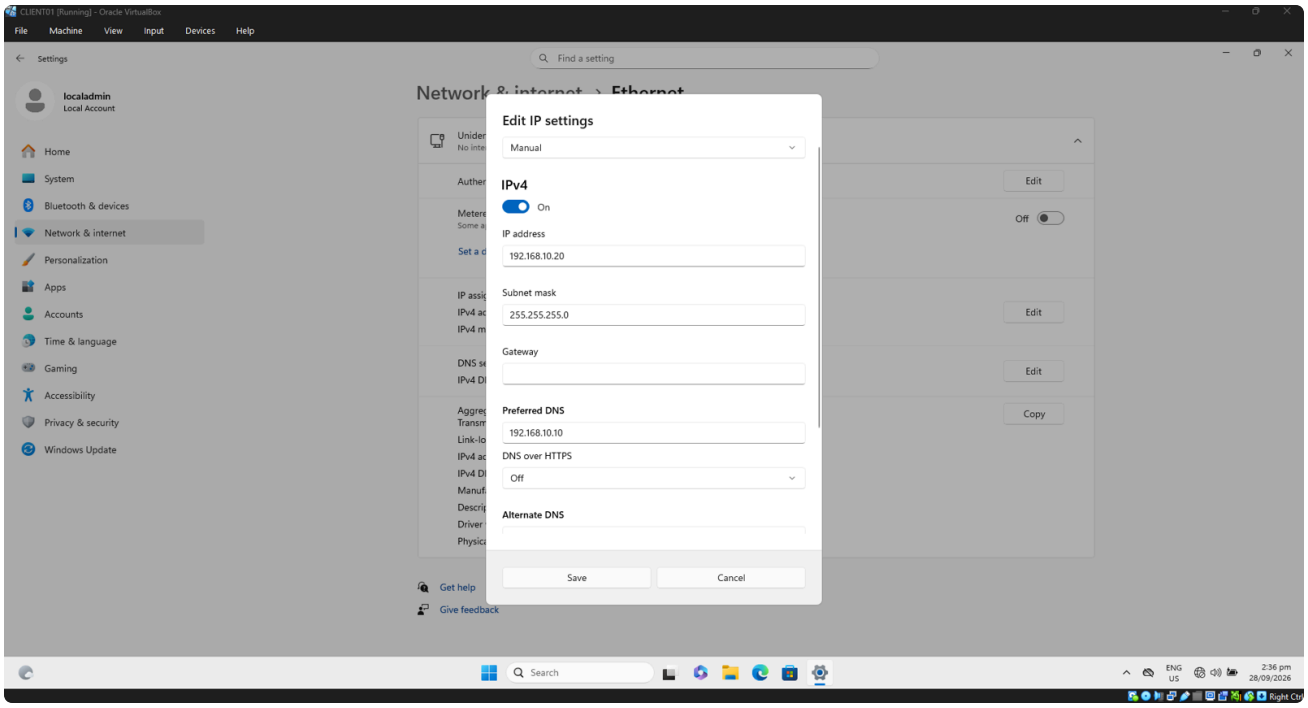


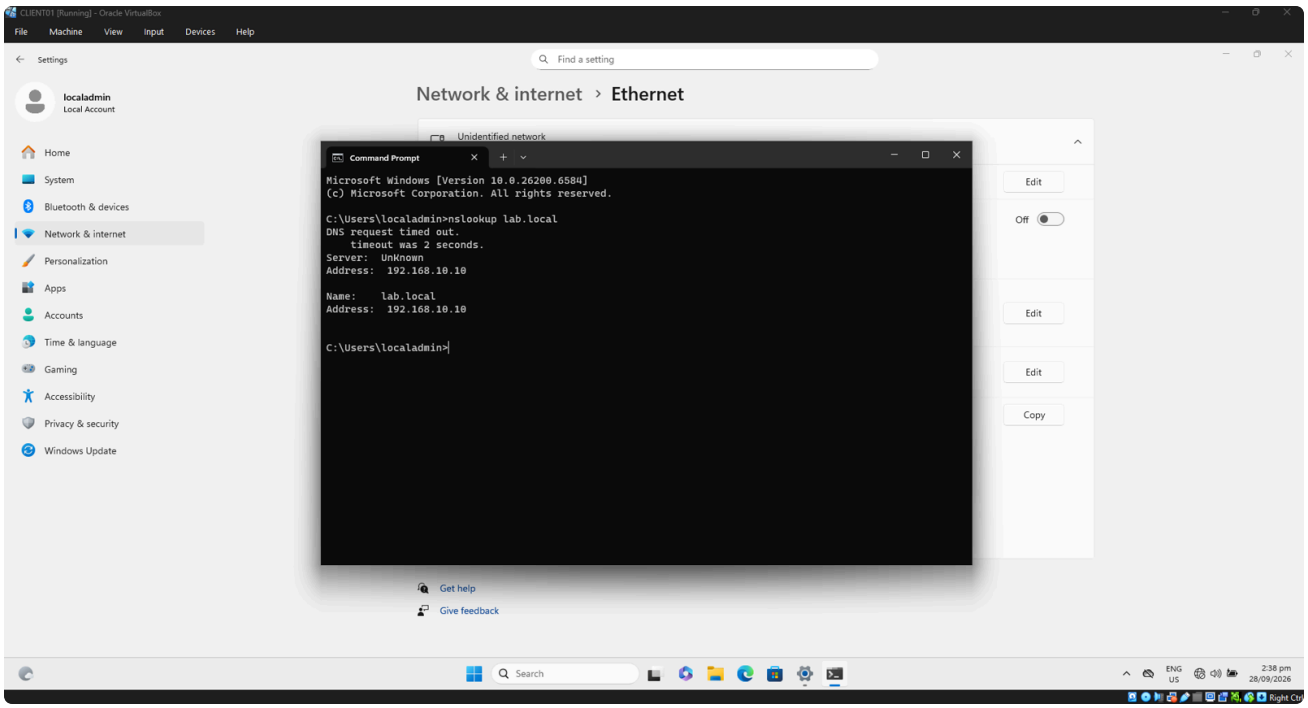


10. **Installed Windows 11 Enterprise** with a local account, `localadmin`. Installed Guest Additions.

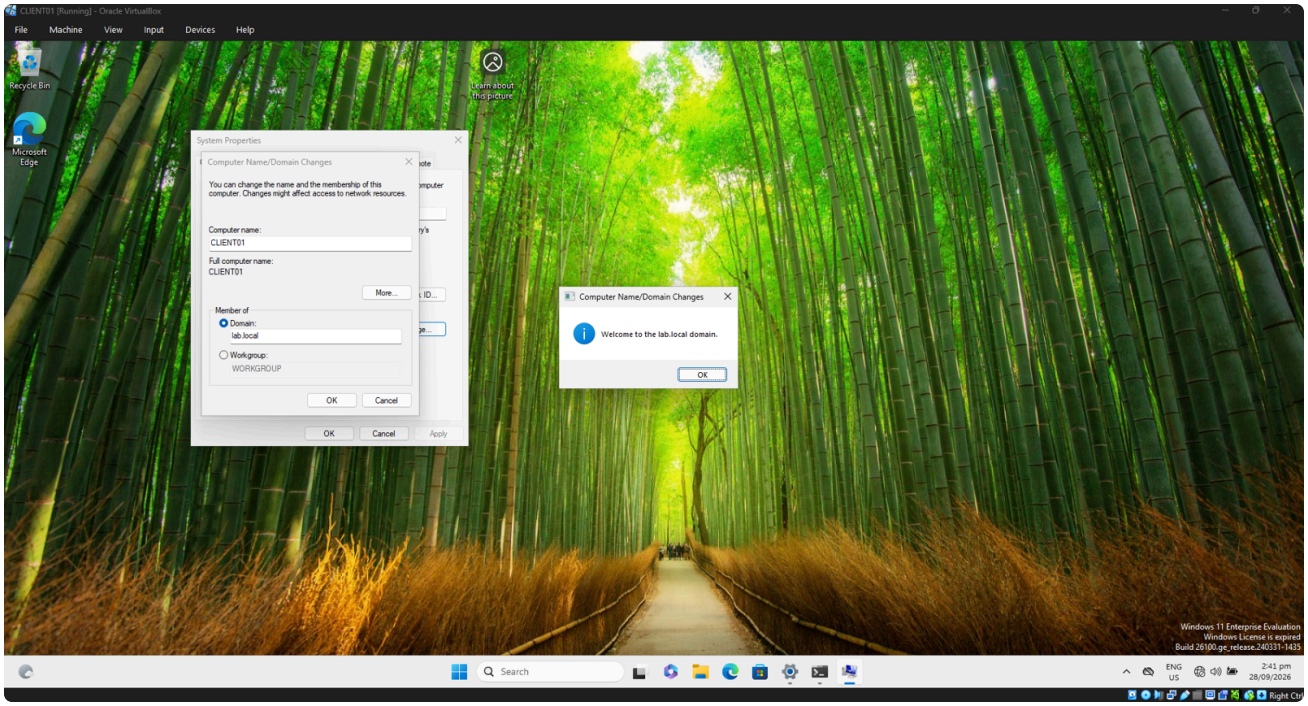


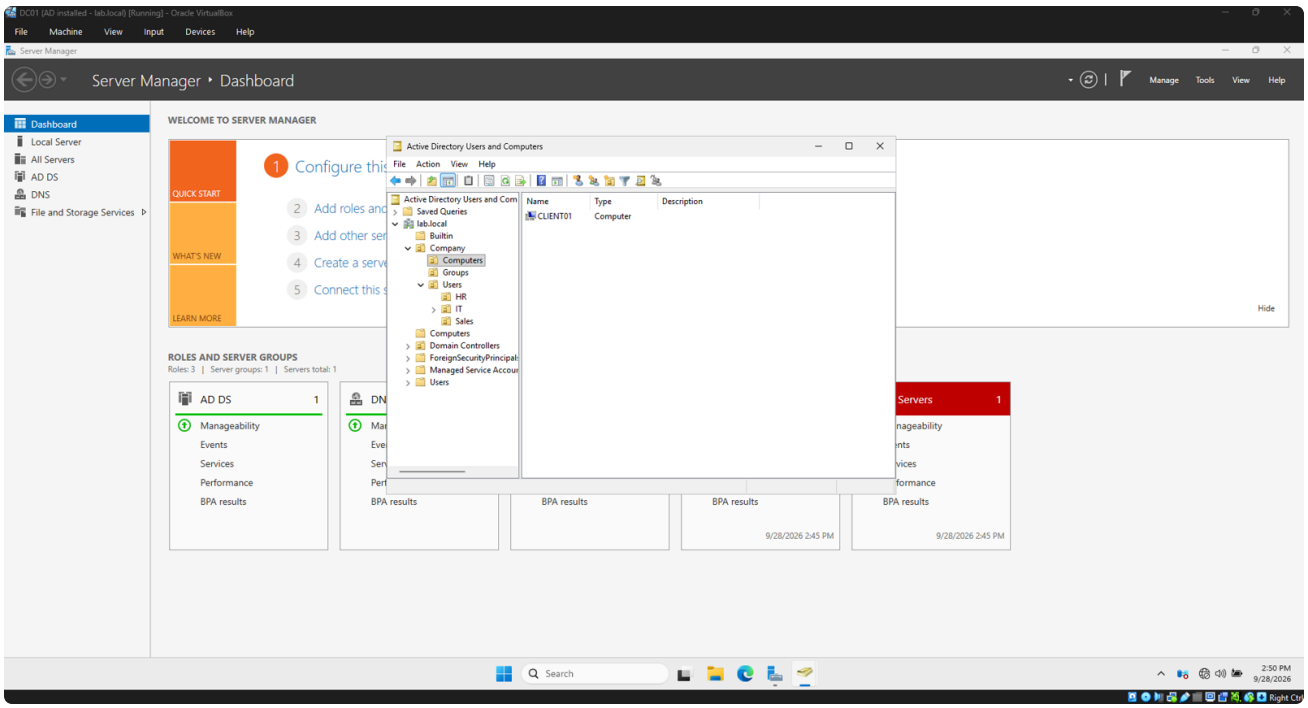
11. **Pointed the client at the domain.** Set a static IP (192.168.10.20/24, DNS 192.168.10.10) and confirmed with `nslookup lab.local`, which returned 192.168.10.10.





12. **Renamed and joined the domain.** Renamed to CLIENT01 and joined `lab.local` using domain Administrator credentials, then restarted. Logged in as `LAB\jsmith`, and moved the CLIENT01 computer object from the default Computers container to `Company\Computers`.





Troubleshooting log

Eight issues came up during the build; seven were fixed and one is known and non-blocking. Each has an error screenshot where one was captured.

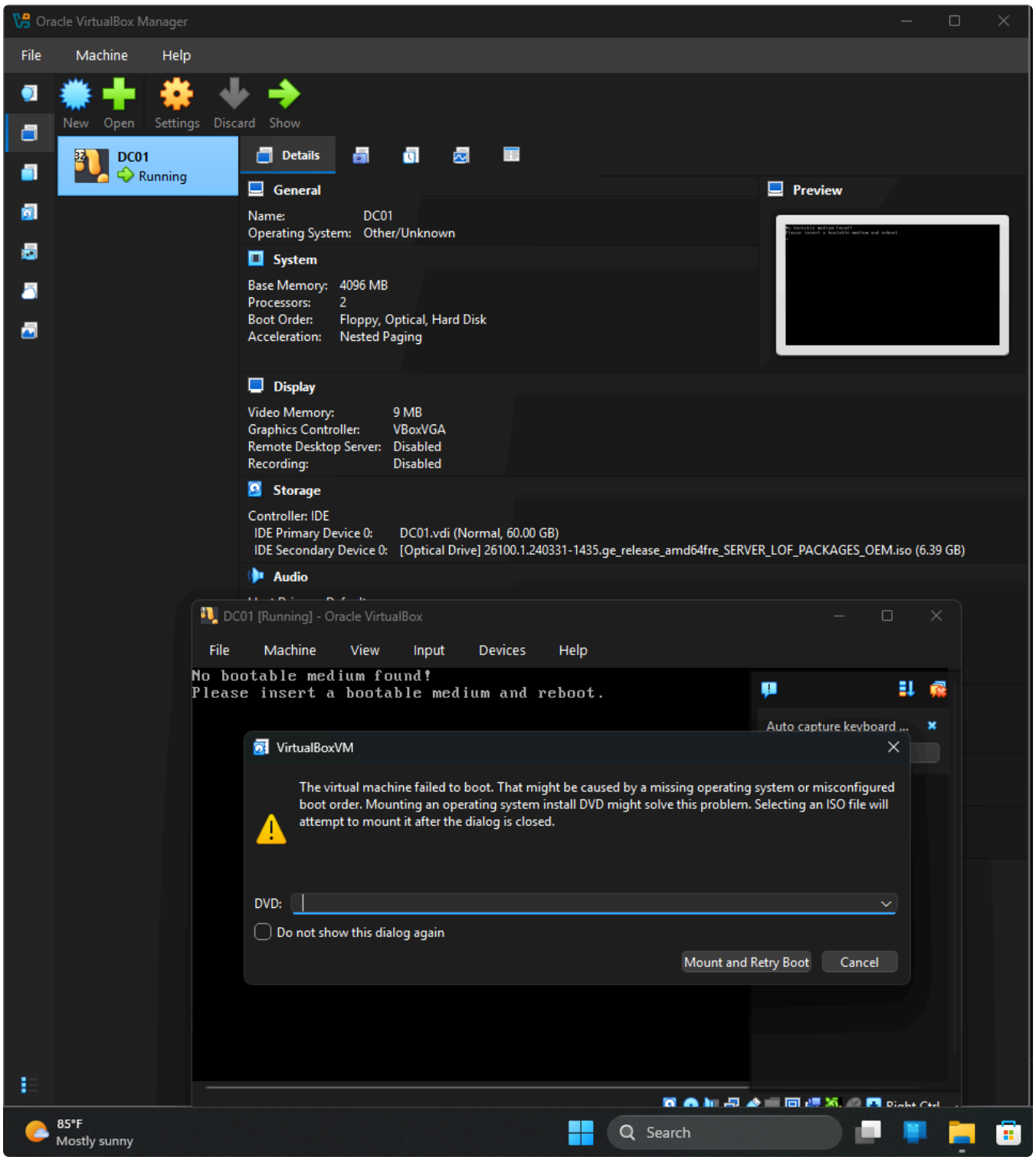
#	Symptom	Cause	Fix
1	DC01 would not boot: "No bootable medium found". VM showed OS type Other/Unknown (32-bit).	Downloaded the wrong ISO: `SERVER_LOF_PACKAGES_OEM.iso` is the Languages and Optional Features disc, not an installer.	Downloaded the 64-bit `SERVER_EVAL` ISO, set the VM type to Microsoft Windows, 64-bit, and remounted it. *Screenshot below.*
2	Ctrl+Alt+Del opened the host's security screen instead of the VM's.	Windows on the host always intercepts Ctrl+Alt+Del.	Sent it with Input > Keyboard > Insert Ctrl-Alt-Del, or Host key (Right Ctrl) + Delete.

#	Symptom	Cause	Fix
3	DC01 clock ran 3 hours ahead of real time.	VM clock not yet synced with the host.	Resolved after installing Guest Additions and rebooting. Matters because Kerberos logins fail when clocks differ by more than 5 minutes.
4	Promotion prerequisites check failed: "Role change is in progress or this computer needs to be restarted."	Started promotion before the AD DS role install had fully finished.	Restarted the server and reran promotion from the Server Manager flag. After the restart, Server Manager briefly showed Roles: 0 until it finished loading. *Screenshot below.*
5	CLIENT01 boot failed: "No bootable option or device was found."	Missed the 5-second "Press any key to boot from CD or DVD" prompt, so the VM tried the empty disk.	Remounted the ISO, restarted, and pressed a key at the prompt. *Screenshot below.*
6	`nslookup lab.local` on CLIENT01 timed out.	DC01 was powered off, so no DNS server answered.	Started DC01 and waited for DNS to load; the lookup then returned 192.168.10.10. The remaining "Server: UnKnown" line is expected with no reverse lookup zone. *Screenshot below.*

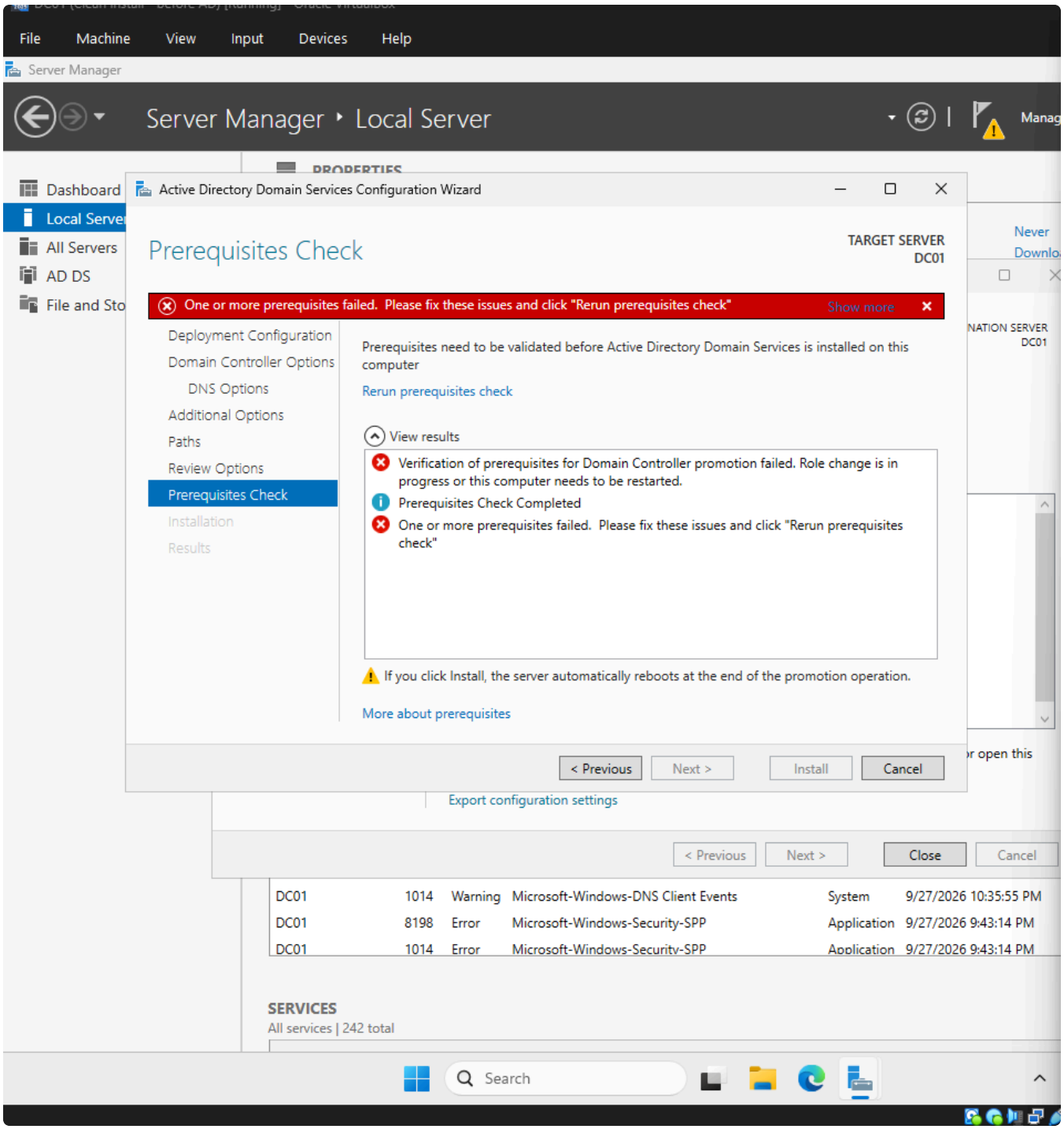
#	Symptom	Cause	Fix
7	After "Welcome to the lab.local domain", an error said changing the Primary Domain DNS name failed.	Known Windows quirk when renaming and joining in one step.	No action needed: the join succeeded. Verified CLIENT01 in ADUC and `whoami` as `lab\jsmith`. *Screenshot below.*
8	CLIENT01 desktop shows "Windows License is expired".	The evaluation cannot activate without internet access.	Open, non-blocking. To revisit if it starts affecting the VM.

Error screenshots

Issue 1: the LOF ISO attached as the boot disc, so DC01 had nothing to boot from.



Issue 4: promotion blocked because the role install hadn't finished.



Issue 5: the boot prompt was missed, so CLIENT01 tried the empty disk.

Image — CLIENT01: No bootable option or device was found

Issue 6: DNS timed out while DC01 was powered off.

Image — nslookup lab.local timing out

Issue 7: the harmless DNS name error shown after a successful join.

Image — Changing the Primary Domain DNS name failed

Verification

Four checks confirm the domain works end to end, from the server side and the client side.

Check	Where	Result
Server is a domain controller	DC01, Server Manager > Local Server	Domain: `lab.local`; AD DS and DNS roles listed; firewall profile Domain: On
Client can find the domain	CLIENT01, `nslookup lab.local`	Resolves to 192.168.10.10
Computer joined the domain	DC01, ADUC	CLIENT01 computer object present, now in `Company\Computers`
Domain user can log in	CLIENT01, `whoami`	`lab\jsmith`

Image — whoami on CLIENT01 returning lab\jsmith

Help desk tickets

With the domain running, I worked simulated help desk tickets end to end: verify the request, fix it, then prove the fix from the user's side.

Ticket	Request	Resolution	Skill
#1001	John Brown (Sales) forgot his password	Temporary password, must change at next logon	Password reset
#1002	New Sales Rep, Maria Garcia, starts Monday	Account in the Sales OU with Sales-Staff access, title, department and manager set	Onboarding
#1003	Jane Smith (HR) locked out	Added a domain lockout policy, then unlocked her account	Lockout policy and unlock
#1004	Alex Lee (IT) leaving today	Account disabled, groups removed, moved to Disabled Users, reason recorded	Offboarding
#1005	Sales needs a private shared folder	`\\DC01\Sales` share, access granted through FS-Sales-Modify (AGDLP)	File shares and NTFS permissions
#1006	Sales keeps forgetting the share's address	GPO linked to the Sales OU maps it as S: at login	Group Policy drive mapping
#1007	nslookup shows "Server: UnKnown"	AD-integrated reverse lookup zone with PTR records	DNS reverse lookups
#1008	Four new starters across Sales and HR	PowerShell script creates accounts from HR's CSV	PowerShell bulk onboarding
#1009	Accounts missing title and department	Found with a `Where-Object` audit, fixed with `Set-ADUser`, re-verified	PowerShell auditing and bulk edits
#1010	Security audit: all admin work done as built-in Administrator	Named `adm-` admin account in its own OU; built-in account kept as break-glass	Privileged account management, UAC

#1001: Password reset

John had forgotten his password after vacation. I reset it in ADUC to a temporary password and ticked **User must change password at next logon**, so IT never knows his real password. In a real environment the caller's identity is verified first, so nobody can reset someone else's password by impersonating them.

Image — Reset Password dialog with must-change ticked

Image — Password for John Brown has been changed

At his next sign-in, Windows forced John to choose a new password before letting him in.

Image — The user's password must be changed before signing in

Image — whoami returning lab\jbrown

#1002: New hire onboarding

Sales manager Karen Mills asked for a new Sales Rep, Maria Garcia, with the same access as the rest of Sales. Karen didn't exist in AD yet, so I created her first, then created Maria (`mgarcia`) in the Sales OU. I added Maria to **Sales-Staff** and set her job title, department (Sales) and manager (Karen Mills). These profile fields feed org charts and address books, and AD fills in Karen's Direct reports automatically.

Image — Maria's Organization tab: title, department, manager

Image — Maria's Member Of tab showing Sales-Staff

Image — whoami returning lab\mgarcia

#1003: Account lockout

Jane reported being locked out, but the domain had no lockout policy, so unlimited wrong guesses were possible. I set one in the **Default Domain Policy**: lock after **5** invalid attempts, for **15 minutes**, with the counter resetting after **15 minutes**. I left Allow Administrator account lockout enabled, since the built-in Administrator is the most-guessed account. After applying it with `gpupdate /force`, I unlocked Jane and confirmed she could sign in. Before unlocking anyone, it's worth asking why the lockout happened: repeat lockouts usually mean an old saved password is retrying somewhere.

Image — Account Lockout Policy: 5 attempts, 15 minutes, 15 minutes

Image — gpupdate /force completing on DC01

Image — whoami returning lab\jsmith after unlock

#1004: Leaver offboarding

HR asked for Alex Lee's access to be removed on his last day, without deleting anything his manager might need. I **disabled** the account rather than deleting it: deletion is permanent and breaks the links to his files and permissions, so companies delete only after a retention period. I removed him from IT-Staff, leaving only Domain Users (his primary group), so the account would grant nothing even if re-enabled by mistake. I then created a **Company\Disabled Users** OU and moved him into it, keeping leavers in one place for auditing.

Image — Alex Lee, disabled, in the Disabled Users OU

Image — Alex's Member Of tab showing only Domain Users

Image — CLIENT01: Your account has been disabled

Alex's account Description was later set to `Disabled 2026-10-01 - Ticket #1004 - left company` (in #1009), so anyone who finds the account knows why it exists and when it was disabled.

#1005: Department file share

Sales manager Karen Mills asked for a shared folder only her team could use. I built it with the **AGDLP** model: **A**ccount → **G**lobal group → **D**omain **L**ocal group → **P**ermission. Maria sits in Sales-Staff (a global group of people), Sales-Staff sits in **FS-Sales-Modify** (a domain local group representing access to the folder), and only FS-Sales-Modify appears on the folder. This separates "who works in Sales" from "who can edit this folder", so future access changes are group-membership edits rather than folder-permission edits, and auditing access means reading one group's members.

The folder `C:\Shares\Sales` is shared as `\\DC01\Sales`. Windows enforces two permission layers and applies the stricter one, so I opened the **share** layer to Authenticated Users (Full Control) and did the real restriction in **NTFS**. Leaving the default share permission (Everyone: Read) would have silently blocked writes. On the folder, I disabled inheritance (converting the inherited entries so Administrators and SYSTEM stayed), removed the inherited **Users** entries that would have let every domain user in, and granted FS-Sales-Modify **Modify** rather than Full Control, so users can't change permissions (least privilege).

Image — FS-Sales-Modify with Sales-Staff as its member

Image — Advanced Security Settings: FS-Sales-Modify has Modify, no Users entries

Testing covered both directions. Maria could create a file, which proves write access over the network, through both layers.

Image — proposal.txt created by Maria in \DC01\Sales

The negative test: Jane (HR) was refused by the NTFS layer, since no entry on the folder matches her groups.

Image — Jane: Windows cannot access \DC01\Sales

#1006: Mapped drive with Group Policy

Sales staff kept forgetting the `\\DC01\Sales` address, so I pushed the share to them as drive **S:** with a Group Policy Object (GPO), a named bundle of settings the domain controller applies to everything in the OU it's linked to.

I created `Sales - Map S Drive` and linked it to the **Company\Users\Sales** OU, so only Sales users receive it; the name states who it targets and what it does. The drive map sits under **User Configuration**, so it follows the person to any PC, and under **Preferences**, the only place drive maps exist (they replaced older `net use` logon scripts). I chose the **Update** action, which creates the drive if missing and corrects it if it exists, unlike Replace, which tears it down at every refresh.

Image — Sales - Map S Drive linked to the Sales OU

Image — Drive map: S: → \DC01\Sales, action Update

The drive appeared for Maria at her next sign-in without running `gpupdate /force`, because Group Policy applies at every login as well as in a background refresh roughly every 90 minutes. Jane, in the HR OU, received no drive, confirming the GPO's scope.

Image — Maria's This PC showing Sales (S:)

Image — Jane's This PC with no S: drive

A drive map is a convenience, not a security control: if it reached Jane by mistake, the NTFS permissions from #1005 would still deny her.

#1007: DNS reverse lookup zone

`nslookup` on the clients reported **Server: UnKnown** and timed out before answering. Forward lookups (name to IP, via A records) worked, but there was no **reverse lookup zone** to turn an IP back into a name (via PTR records), so nslookup couldn't name the DNS server it was talking to. Reverse lookups also matter for logs and security tools that record only IP addresses.

I created a primary, **AD-integrated** reverse zone, so its records replicate to future domain controllers automatically. I set replication to all DNS servers in the domain and allowed **secure dynamic updates only**, so only authenticated domain computers can register or change their own records, and a rogue device can't claim a server's name.

My first attempt failed: I typed the network ID as `192.68.10` instead of `192.168.10`, creating the zone `10.68.192.in-addr.arpa`. Creating DC01's PTR record then failed with "the referenced reverse lookup zone cannot be found", because DNS looked for `10.168.192.in-addr.arpa`. Zones can't be renamed, so I deleted the zone and recreated it, this time checking the zone name on the wizard's summary page before finishing.

Image — PTR creation fails: the zone was misnamed 10.68.192.in-addr.arpa

With the zone correct, I created PTR records for DC01 and CLIENT01 from their existing A records, which guarantees the name and IP match.

Image — 10.168.192.in-addr.arpa with PTR records for DC01 and CLIENT01

`nslookup lab.local` now names the server as dc01.lab.local, and a direct reverse lookup of 192.168.10.20 returns CLIENT01.lab.local.

Image — nslookup showing dc01.lab.local and resolving 192.168.10.20 to CLIENT01

#1008: Bulk onboarding with PowerShell

HR sent four new starters across Sales and HR. Rather than creating them by hand, I scripted it, so the same steps scale to any number of people, every field is filled identically, and the script itself records what was done.

I saved HR's list as `C:\Scripts\newhires.csv`, adding a Username column following the first-initial-plus-surname convention, and checked it with `Import-Csv` before changing anything. The script below loops over each row, builds the target OU's distinguished name from the Department column, creates the account with splatted parameters, and adds it to the matching `<Department>-Staff` group, which only works because group names follow a consistent convention. The temporary password is prompted for at run time and never stored in the file. `New-ADUser` creates accounts disabled by default, so the script sets `Enabled` explicitly.

```
$password = Read-Host "Temporary password for new starters" -AsSecureString
$users = Import-Csv "C:\Scripts\newhires.csv"

foreach ($u in $users) {
    $ou = "OU=$($u.Department),OU=Users,OU=Company,DC=lab,DC=local"

    $params = @{
        Name                = "$($u.FirstName) $($u.LastName)"
        GivenName           = $u.FirstName
        Surname             = $u.LastName
        DisplayName         = "$($u.FirstName) $($u.LastName)"
        SamAccountName      = $u.Username
        UserPrincipalName   = "$($u.Username)@lab.local"
        Title               = $u.Title
        Department          = $u.Department
        Path                = $ou
        AccountPassword     = $password
        ChangePasswordAtLogon = $true
        Enabled             = $true
    }

    New-ADUser @params
    Add-ADGroupMember -Identity "$($u.Department)-Staff" -Members $u.Username
    Write-Host "Created $($u.Username) in $($u.Department)"
}
```

Image — Script output: four accounts created

A `Get-ADUser` audit confirmed each account's title and department, and `Get-ADGroupMember` confirmed the group memberships. The audit also exposed that the two earliest hand-made accounts, Jane Smith and John Brown, had no title or department, which is the kind of inconsistency PowerShell audits catch and manual clicking misses.

Image — Get-ADUser and Get-ADGroupMember verification

The real test was Priya Patel's first login. Without any per-user setup, she received the S: drive (from the Sales OU's GPO, #1006) and could open it (through Sales-Staff → FS-Sales-Modify, #1005). Once the structure is right, onboarding is just creating the account in the right OU and group.

Image — Priya: whoami lab\ppatel with the S: drive open

#1009: Auditing and fixing account data

The #1008 audit showed accounts with no title or department, which makes people vanish from HR reports and the address book. Rather than fixing the two names I already knew, I searched for the condition itself, so any other incomplete account would surface too:

```
Get-ADUser -Filter * -SearchBase "OU=Users,OU=Company,DC=lab,DC=local" -
Properties Title,Department |
    Where-Object { -not $_.Title -or -not $_.Department } |
    Select-Object Name,SamAccountName,Title,Department
```

It returned Jane Smith and John Brown. My first fix contained a mistake: the second `Set-ADUser` targeted `jsmith` instead of `jbrown`, overwriting Jane's new values and leaving John blank. Rerunning the same search is the step that catches this kind of error. I corrected both accounts, and the rerun search returned nothing. The mistake changed only a descriptive field; access was never affected, because it comes from group membership and OU placement, not profile attributes. In the same session I recorded why Alex Lee's account is disabled (#1004).

Image — The audit, the mistaken fix, the correction and the empty rerun

Lesson: a check only finds what it's designed to look for. The blank-field search proves nothing is missing, but not that values are correct; reading changed accounts back covers that.

#1010: Named admin accounts

A simulated security audit found all admin work being done as the built-in **Administrator**. That account is shared (so logs can't say which person acted), is the first account attackers target, and by default bypasses UAC, so everything it runs has full rights.

I created a **Company\Admin Accounts** OU, kept beside Company\Users rather than inside it so user-targeted GPOs never reach admin accounts, and a named admin account, `adm-zramkissoon`, added to Domain Admins. The `adm-` prefix makes privileged accounts obvious in logs and easy to find with a script. The built-in Administrator stays enabled as a **break-glass** account for emergencies only.

My first `New-ADUser` attempt failed. Pasting lost the line breaks, so the line-continuation backticks landed mid-line, where a backtick instead escapes the next character, turning `-SamAccountName` into plain text ("A positional parameter cannot be found"). The OU path also had `OU-` instead of `OU=`. I reran it as a single line with no backticks.

Image — New-ADUser failing: backticks mid-line after pasting

Signed in as the new account, `whoami /groups` showed UAC at work. In a normal window, Domain Admins is present but marked **Group used for deny only**, so its rights are inactive.

Image — Normal window: Domain Admins is deny-only

In an elevated window, the same group is **Enabled**. Admin rights apply only where I deliberately elevate.

Image — Elevated window: Domain Admins is enabled

The group's SID ends in **512**, the fixed identifier for Domain Admins in every domain; the built-in Administrator is always **500**, which is part of why it's such a predictable target.

Linux server build: HELPDESK01

HELPDESK01 is an Ubuntu Server 26.04 LTS VM at 192.168.10.30 that will host the osTicket help desk, managed over SSH from CLIENT01. I chose an LTS release for five years of security updates, and the Server edition without a desktop, which runs in 2 GB of RAM and is managed the way real Linux servers are.

Setting	Value	Why
vCPU / RAM / disk	2 / 2 GB / 25 GB	Enough for a web server and database without a GUI
Adapter 1	NAT, DHCP (10.0.2.15)	Outbound-only internet for package downloads; nothing can reach in
Adapter 2	labnet, static 192.168.10.30/24, no gateway	Fixed address on the lab network; one default route, via NAT
DNS	192.168.10.10, search domain `lab.local`	Split DNS: lab names resolve via DC01, everything else via NAT
Storage	LVM, whole disk	Volumes can be grown later without reinstalling
Remote access	OpenSSH server	Encrypted remote administration
Admin account	Local user with sudo	Elevate per command instead of logging in as root

Image — Adapter 1: NAT

Image — Adapter 2: Internal Network labnet

The first install attempt hung for nearly an hour at "Loading essential drivers." The Ubuntu installer runs entirely from RAM, so I raised the VM to 4 GB for the install only, and disabled 3D acceleration, a common cause of Linux boot hangs in VirtualBox. The second attempt booted normally. During installation the kernel logged a "soft lockup" warning, meaning a virtual CPU was starved of host CPU time for 39 seconds; it was non-fatal and pointed to a busy host rather than a fault. After installing, I patched the system and returned the RAM to 2 GB.

Image — SSH from CLIENT01 to helpdesk01, fully patched

The SSH session from CLIENT01 confirms the static lab address works, the host key was verified and recorded on first connection, and no updates are outstanding.

osTicket help desk on a LAMP stack

osTicket 1.18.4 now runs on HELPDESK01 at `http://helpdesk01.lab.local`, with the staff panel at `/scp/`. It sits on a LAMP stack: Linux, Apache, MariaDB and PHP.

Compatibility check first. osTicket 1.18.4 supports PHP 8.2 to 8.4, but Ubuntu 26.04 ships PHP 8.5. Rather than install an unsupported version, I added the signed packages.sury.org repository, with its key scoped to that repository only via `Signed-By`, and installed PHP 8.4 explicitly, naming `php8.4-*` packages so Ubuntu's default 8.5 was never pulled in.

Image — apt-cache policy: PHP 8.4 candidate from packages.sury.org

Image — PHP 8.4.26; Apache and MariaDB active and enabled at boot

Database, least privilege. I ran `mariadb-secure-installation` (removing anonymous users, remote root login and the test database; root stays on Unix-socket authentication, so it has no password to steal) and created a dedicated `osticket` user that can connect only from localhost and only reach the `osticket` database. Logged in as that user, only `osticket` is visible.

Image — The osticket database user sees only its own database

Web server. osTicket's files live in `/var/www/osticket`, served by its own Apache virtual host with separate error and access logs. The config was validated with `apache2ctl configtest` before reloading. I added a DNS A record and PTR on DC01, so users reach the site by name.

Image — apache2ctl configtest: Syntax OK

Image — helpdesk01.lab.local resolves to 192.168.10.30 via DC01

Install and lockdown. The installer's prerequisite check passed every required item; only IMAP (email fetching, unused here) was missing. During install, Apache's account (`www-data`) owned the files so it could write the config. Afterwards I deleted the `setup` folder, so nobody can rerun the installer, and made `ost-config.php`, which holds the database password, owned by root and read-only to Apache (`0644`).

Image — osTicket prerequisites check

Image — osTicket installation complete

Image — Staff Control Panel with the first ticket

Key concepts and next steps

The biggest lesson from Part 1: AD depends on DNS, so most join and login problems start there.

Concepts put into practice

- **Domain controller:** stores the directory and authenticates logins. It also runs DNS, so clients must use its IP as their DNS server.
- **OUs vs groups:** an account lives in exactly one OU, used for organizing, Group Policy and delegation. It can belong to many groups, used for granting access.

- **Rename before promoting:** renaming a domain controller after promotion is disruptive, so DC01 was named first.
- **Default containers:** new computers land in the built-in Computers container, where GPOs cannot be linked, so they're moved into a managed OU.
- **Snapshots:** taken at each milestone so the lab can be rolled back in seconds.

Roadmap

The lab grows from on-prem Active Directory into a small, connected IT environment: a Linux server running a real ticketing system, hybrid cloud identity, and an AI help desk assistant. Each phase builds on the AD foundation rather than standing alone.

Phase	Focus	How it ties into the lab	Status
1	Core AD build	Domain controller, `lab.local`, OUs, users, groups, domain-joined client	Done
2	Help desk tickets #1001–#1009	Password resets, onboarding and offboarding, lockout policy, file shares, Group Policy, DNS, PowerShell	Done
3	Admin account separation (#1010)	Named `adm-` admin accounts; built-in Administrator kept for emergencies	Done
4	Linux server and osTicket	Ubuntu Server VM hosting osTicket, users log in with AD accounts, server joined to the domain	In progress
5	Remaining AD work, ticketed in osTicket	Delegation, DHCP, Group Policy troubleshooting, security filtering, AD Recycle Bin, second DC and replication, FSMO roles, backup and restore	Planned
6	Hybrid identity with Microsoft Entra ID	Lab users synced to the cloud for on-prem and cloud sign-in	Planned

Phase	Focus	How it ties into the lab	Status
7	AI help desk assistant	Agent triages osTicket tickets and proposes fixes; least-privilege delegated account, human approval, full logging	Planned

The AI assistant comes last on purpose: it depends on the ticketing system's API, delegated permissions from phase 5, and PowerShell automation.